

Requirements Table : Peer-to-Peer Carpooling & Expense Splitter

Problem Statement #26 | Actors: Rider Commuter, Driver Host

Req - ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate route compatibility between driver and passenger commute paths, permitting matches only within a 1.5 km detour tolerance.	High	Pass: Route match confirmed and fuel cost split calculated. Fail: Detour exceeding tolerance accepted.	Core matching logic; prevents drivers from being routed excessively out of their way.
FR-002	Functional	The system shall allow a Rider Commuter to confirm a proposed match and book a seat on the Driver Host's trip.	High	Pass: Seat is reserved and both parties receive a booking confirmation within 5 seconds of confirmation. Fail: Booking accepted without reserving a seat or without notifying both parties.	Booking is the trigger that converts a match into a committed shared trip.
FR-003	Functional	The system shall automatically calculate each rider's fuel cost share based on distance travelled, current fuel price, and number of co-riders on the trip.	High	Pass: For a given trip, the sum of all riders' cost shares equals the total trip fuel cost (+/- Rs.1 rounding). Fail: Cost shares do not sum to total trip cost or are not shown before payment.	Automated, transparent splitting is the core value proposition of the platform.
FR-004	Functional	The system shall verify a Driver Host's driving license and vehicle registration documents before allowing them to publish a ride offer.	High	Pass: A driver whose documents fail verification cannot publish a ride offer and sees a rejection reason. Fail: An unverified driver is able to publish a ride offer.	Protects riders' safety and platform trust; required before any driver-side matching can occur.
FR-005	Functional	The system shall allow Rider Commuters and Driver Hosts to rate each other and leave feedback after a trip is completed.	Medium	Pass: A rating prompt appears within 1 minute of trip completion and a submitted rating (1-5) is stored against both user profiles. Fail: Rating prompt does not appear or rating is not saved.	Ratings build trust in the peer-to-peer network and inform future match quality.
NFR-001	Non-functional	User ratings, driving license verification documents, and travel histories must be stored securely with encrypted backups.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Sensitive personal and identity data requires strong protection to maintain user trust and comply with data-protection norms.
NFR-002	Non-functional	The system shall support at least 5,000 concurrent route-matching requests during peak commute hours (8-10 AM, 5-7 PM) while maintaining a 95th-percentile match response time under 3 seconds.	Medium	Pass: Load test with 5,000 simulated concurrent users shows p95 match response time <= 3s and zero failed requests. Fail: Response time exceeds 3s or requests are dropped under peak load.	Commuting is inherently time-bound and concentrated in narrow windows, so the system must remain responsive exactly when demand spikes.